# Xingyi (Daniel) Chen

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## RESEARCH INTERESTS

Statistical genomics and biomedical data science; development and evaluation of statistical methods for high-dimensional biomedical data.

## EDUCATION

**Johns Hopkins University** Baltimore, MD

*B.S. in Applied Mathematics & Statistics, minors in Computational Medicine and Mathematics* Expected May 2027

- GPA: 3.99/4.00 | Dean's List, 6 semesters
- Relevant Coursework:
  **Mathematics & Statistics:** Real Analysis I/II (with Measure Theory), Probability, Mathematical Statistics, Applied Statistics & Data Analysis, Monte Carlo Methods, Optimization, Computational Mathematics (*in progress*), Multivariable Calculus, Linear Algebra, Differential Equations, Causal Inference (*audit*)
  **Computational Biology & Genomics:** Computational Genomics: Data Analysis, Computational Biology & Bioinformatics, Algorithms for Genomic Sequence Data (*in progress*), Genomic Data Visualization, Computational Medicine, Systems Pharmacology, Translating Healthcare AI (*in progress*)

## PUBLICATIONS & PREPRINTS

**Chen X**, Hicks SC. Learning interpretable compositional features associated with continuous covariates. *bioRxiv*, 2026. https://doi.org/10.64898/2026.11.04.764349

**Chen X**, Li J, Battle A. Evaluation design shapes estimates of cross-donor generalization in single-cell classification of cell-type. *bioRxiv*, 2026. https://doi.org/10.64898/2026.10.26.761949

**Chen X**, Totty M, Hicks SC. SpotSweeper-py - spatially-aware quality control metrics for spatial omics data in the Python ecosystem. *F1000Research*, 2026; 15:33. https://f1000research.com/articles/15-33/v2

## RESEARCH EXPERIENCE

**Research Assistant – Hicks Lab** January 2025 – Present

Johns Hopkins University, Department of Biostatistics | PI: Stephanie Hicks, PhD

- Developed *IsoVita*, a statistical framework that learns interpretable features from compositional data and tests their associations with continuous covariates; designed complementary tests for abrupt and gradual changes with permutation inference accounting for data-driven feature learning.
- Evaluated false discovery control, detection power, and change point accuracy through simulations; applied *IsoVita* to hippocampal and dorsolateral prefrontal cortex RNA-seq data to identify age-associated isoform usage patterns and released an open-source Python package.
- Developed and released *SpotSweeper-py* for spatial omics quality control in Python/scverse; validated agreement with the R/Bioconductor implementation and benchmarked computational performance and parameter sensitivity.
- Built reusable Visium HD spatial transcriptomics analysis infrastructure, including 10+ reproducible pipelines and the *visiumhd-utils* Python package for preprocessing, QC, destriping, visualization, and figure generation; supported an NIH R01 grant proposal.

**Quantitative Sciences Undergraduate Research Experience (QSURE) Intern – Shah/McPherson Lab** New York, NY

Memorial Sloan Kettering Cancer Center | PI: Sohrab Shah, PhD; Mentors: Andrew McPherson, Matt Myers Summer 2026

- Developed a deletion-specific one-sided scan statistic for low-coverage scDNA-seq, extending a multi-sample scan framework to detect homozygous deletions by comparing copy number zero with the best-fitting nonzero state.
- Benchmarked CBS, PELT, FPOP, dynamic programming segmentation, and scan-statistic approaches across sequencing depth, bin size, event length, aneuploidy, and subclonal prevalence, identifying failure modes and resolution limits.
- Calibrated the statistic using empirical null distributions and assessed robustness, characterizing recovery of rare subclonal deletions and sensitivity to bin alignment, carrier prevalence, and noise estimation.

- Built an end-to-end Python workflow for preprocessing, interval scanning, posterior carrier scoring, parallel computation, and visualization; applied it to MSK-SPECTRUM ovarian cancer scDNA-seq data to investigate candidate deletions and technical artifacts.

**Independent Research Project – Cross-donor Generalization in scRNA-seq** March 2026 – Present
Johns Hopkins University, Department of Biomedical Engineering | PI: Alexis Battle, PhD

- Led a multi-cohort study of scRNA-seq cell-type classification, comparing random cell-level, donor-held-out, and cross-site evaluation across HVG, PCA, Harmony, and scVI representations.
- Assessed evaluation gaps using class-specific analyses, batch covariate adjustment, and donor-level mixed-effects models; found that random splits overestimated cross-donor generalization performance in PBMC and lung datasets, while gaps were minimal in near-ceiling benchmarks and a randomized-label control.
- Built a modular, reproducible Python benchmarking framework for multi-cohort analysis, enabling systematic comparison of evaluation designs and expression representations.

**Research Assistant – Beer Lab** November 2023 – January 2025
Johns Hopkins University, Department of Biomedical Engineering | PI: Michael Beer, PhD

- Automated extraction and preprocessing of gene expression datasets from public databases and performed exploratory analyses and visualizations in R using tidyverse and ggplot2.

**Bioinformatics Intern – Wei Lab** Shanghai, China
Lin Gang Laboratory | PI: Wu Wei, PhD Summer 2024

- Reviewed and presented methodological summaries of five spatial-omics studies, highlighting statistics/machine learning approaches and potential follow-up analyses; supported two ongoing bioinformatics projects with data cleaning and exploratory analyses.

## PRESENTATIONS

**Chen X**, Myers M, McPherson A. *Identifying homozygous deletions in single-cell DNA sequencing data*. Oral presentation at *QSURE Symposium*, Department of Epidemiology & Biostatistics, Memorial Sloan Kettering Cancer Center, August 2026.

**Chen X**, Erdogdu B, Pertea M, Hicks SC. *Machine learning reveals tissue-agnostic and region-specific isoform aging markers in the human hippocampus.* Poster presentation at *Biology of Genomes*, Cold Spring Harbor Laboratory, May 2026.

## TEACHING & LEADERSHIP EXPERIENCE

**Undergraduate Lead Teaching Assistant** August 2024 – Present
Johns Hopkins University, Department of Mathematics | Supported Course: Differential Equations & Applications

- Promoted from Course Assistant to Lead TA; independently prepared and led 13 weekly discussion sections per semester for 25 students.
- Mentored undergraduate TAs, coordinated course logistics, co-developed quizzes/review sessions, and earned an average student teaching-evaluation score of 98%.

**Co-Instructor & Peer Leader** August 2026 – Present
Johns Hopkins University | Supported Course: Intro to Hopkins: Arrive & Thrive

- Co-instructed a first-year seminar introducing incoming students to academic, professional, and student-support resources at Johns Hopkins.
- Mentored first-year students through their transition to university and facilitated discussions on academic success, community, and campus engagement.

## SKILLS

**Programming:** Python, R, Bash
**Scientific Computing:** Scanpy, AnnData, scikit-learn, Bioconductor; Linux, HPC/Slurm
**Development & Reporting:** Git/GitHub, Jupyter, Quarto, LaTeX